

EXTERNAL INTELLIGENCE FOR IT OPERATIONS

InfraRisk Analyzer

Review infrastructure changes before they become outages.

InfraRisk Analyzer extends the Assignment 1 Change Review Board with request-time External Intelligence. An Operator selects a locally stored IT Change, searches NVD by vendor/product, or enters a known CVE. The application retrieves technical evidence from the NVD JSON API and an exploitation/remediation signal from the CISA Known Exploited Vulnerabilities catalog.

Boundary: External evidence informs human review. It does not prove that the selected Affected System is vulnerable, change the manually assigned Risk Level, approve or block the IT Change, or create a persistent source record.

API source

NVD API 2.0 returns the English description, publication date, best CVSS metric, and record URL.

Scraped source

CISA's public KEV table contributes vendor, product, vulnerability, dates, and required action.

Unified purpose

Both sources answer what current vulnerability evidence an Operator should consider before a change window.

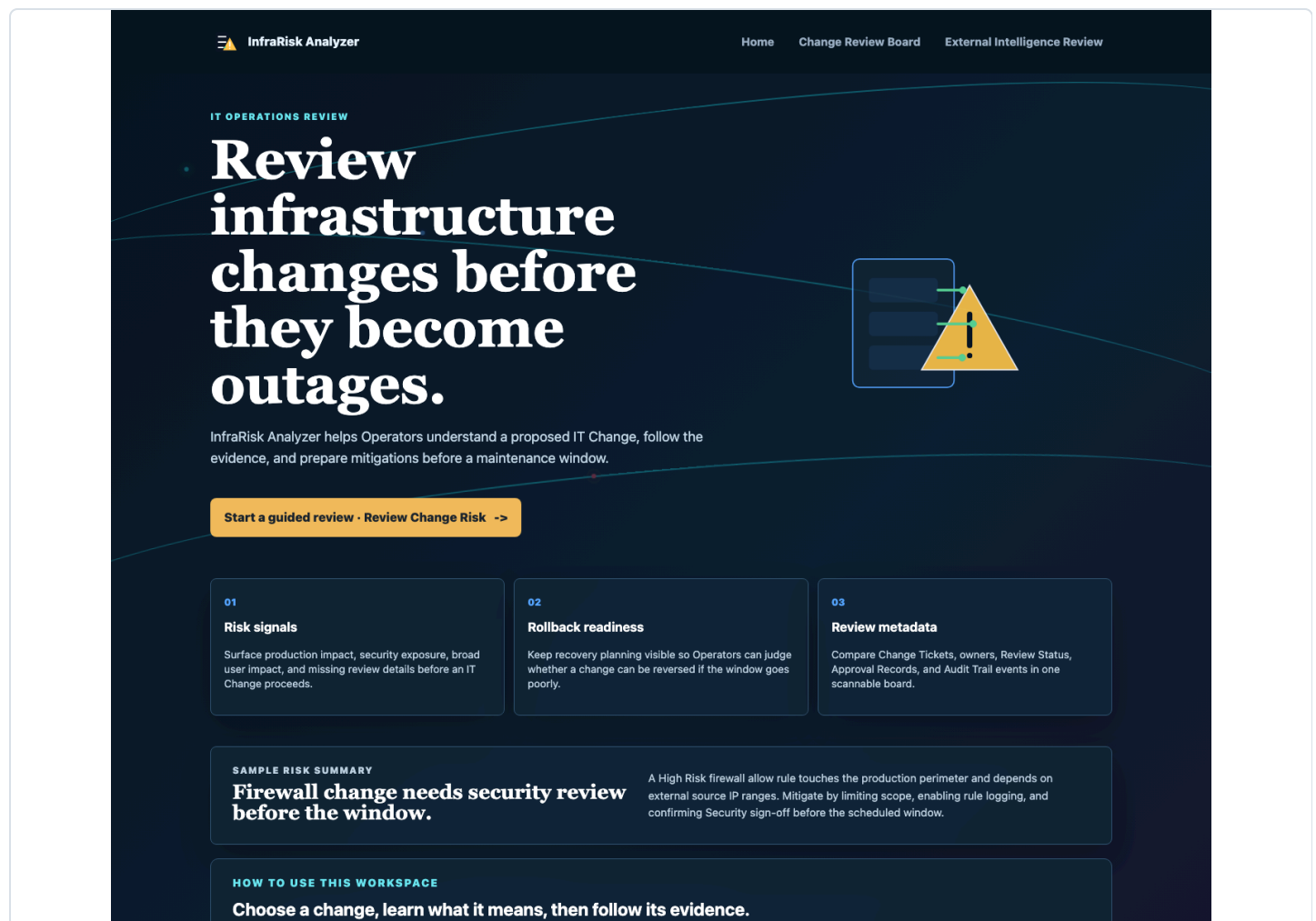
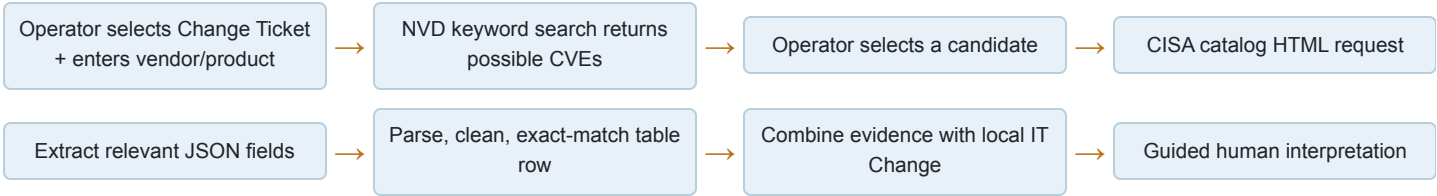


Figure 1. Final homepage: a human-centered command-center visual system with guided review entry.

One review, two external evidence boundaries



Source and environment contract

Source	Endpoint / fields	Purpose and boundary
NVD API 2.0	<code>https://services.nvd.nist.gov/rest/json/cves/2.0</code> English description, published, CVSS, NVD URL	Technical vulnerability record; uses <code>NVD_API_KEY</code> , a normalized <code>cveId</code> , User-Agent, and a finite timeout.
CISA KEV catalog	<code>https://www.cisa.gov/known-exploited-vulnerabilities-catalog</code> Vendor/project, product, name, dates, required action	Public webpage scrape; exact CVE match only. Fifteen-minute process-local cache reduces repeat requests.

Environment variables: `NVD_API_KEY` is required for NVD; `NVD_API_URL` , `CISA_KEV_URL` , and `SECRET_KEY` are optional overrides/configuration. A placeholder-only `.env.example` is committed; real `.env` is ignored.

Information model

Local IT Change

title, Change Type, Risk Level, Affected System, window, Risk Signals, Mitigation, Rollback Plan, Change Owner, Change Ticket, Review Status, Approval Records, Audit Trail.

Request-time evidence

NVD: CVE Identifier, description, published, CVSS, source URL, retrieval time.

CISA: CVE Identifier, vendor/project, product, vulnerability name, dates, required action, source URL, retrieval time.

External Intelligence Review

The selected Change Ticket remains visible beside both source panels. Guided questions separate source facts from Operator interpretation and preserve the local manually assigned Risk Level.



Figure 2. Final Change Review Board: compact interactive cards, expandable review detail, and full-page data highways.

Failure-handling matrix

Condition	Behavior	Test evidence
Invalid CVE or unknown Change Ticket	Rejects before acquisition with a plain-language 400/404 page; no outbound request for invalid CVE.	Route parametrized tests.
Missing NVD key	503 configuration message; NVD request is not called.	<code>test...missing_key...</code>
NVD failure, rate limit, network, malformed JSON	502 source-unavailable message, or 429 retry-later message; no traceback.	HTTP, timeout, and JSON tests.
Empty NVD result	404 "No NVD record found" message.	Empty-result route test.
Changed/missing CISA markup or network failure	503 partial state preserves NVD evidence and says CISA listing status cannot be determined.	Scraper malformed-row/header and partial-result tests.
CISA exact-match miss	Displays "not listed in this catalog snapshot"; never claims safe or non-exposed.	No-match route test.

Verification record

- Automated suite: **43 passed** with mocked sources; no live credential is required for reproducibility.
- Production routes verified in the checked-in screenshots: homepage, compact Change Review Board, External Intelligence Review, side-by-side discovery layout, source details, guided questions, and narrow responsive layout.
- Reduced-motion behavior is implemented in `static/style.css`; JavaScript enhancement keeps all answers available without scripting.
- Repository scan: no tracked API key, token, password, local environment file, or secret-bearing config; only placeholder `.env.example`.
- Live check: authenticated NVD keyword search returned normalized results; the current CISA card markup also parsed successfully for a current catalog entry.

Ethics, limitations, and Assignment 3

Ethical considerations

- Attributes NVD and CISA and links to their source records.
- Uses the documented NVD endpoint, API key header, descriptive User-Agent, finite timeout, and respectful request frequency.
- Scrapes only public CISA table fields needed for review; respects source terms, licensing, robots guidance, and layout changes.
- Does not collect personal information or infer compromise from a CVE match.

Known limitations

- Evidence is request-time and process-local CISA caching is lost on restart.
- External schemas, availability, rate limits, and HTML layout can change.
- A CVE does not establish that the selected system uses the affected product/version.
- Demonstration data is not a production approval system.

Assignment 3 direction

Add persistent IT Changes, Risk Findings, Approval Records, Audit Trail events, and timestamped External Intelligence Snapshots. A snapshot will retain the normalized NVD/CISA evidence, source URLs, retrieval timestamps, selected Change Ticket, and Operator review context without turning external evidence into automatic approval or blocking logic.

Release checklist

The repository contains the A1 PDF unchanged plus this versioned A2 PDF, updated README sections, focused service and route tests, NVD discovery, CISA table/card parsers, and checked-in browser screenshots. Final verification: `./venv/bin/python -m pytest -q` reports 43 passing tests. A private-key live NVD check and a current CISA card-parser check succeeded without exposing the credential; an older requested CVE was correctly reported as unavailable when it was not present in the current CISA result page.